How-To Library

Informatica

Deploy Axon Data Governance 7.2.3 on the Google Cloud Marketplace

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Abstract

The automated marketplace solution uses Google Cloud Platform templates to launch, configure, and run the Google Cloud compute, network, storage, and other services required to deploy a specific workload on Google Cloud. This deployment reference provides step-by-step instructions for deploying Informatica® Axon Data Governance on Google Cloud Platform from Google Marketplace.

Supported Versions

- Axon Data Governance 7.2.3

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Apache Log4j RCE Vulnerability for Axon Data Governance

The Log4j vulnerability is addressed in the following ways:

- The Elasticsearch service in Axon is updated to version 6.8.22, which uses Log4j version 2.17.0.
- The Axon Agent uses Log4j version 2.17.1.

Overview

This deployment article provides step-by-step instructions to deploy Axon Data Governance on the Google Cloud Platform. Automated reference deployments use Google Cloud templates to launch, configure, and run the Google Cloud compute, network, storage, and other services required to deploy a specific workload on the Google Cloud Platform.

Intended Audience

This article is intended for users who are responsible for deploying Axon Data Governance 7.2.3 on the Google Cloud marketplace.

As a user with administrator privileges to deploy applications on the Google Cloud Platform, we assume that you are familiar with the following Google Cloud resources:

- Google deployment manager
- Network on the Google cloud
- Compute Engine
- Google Storage
- Firewall Rules
- Cloud NAT
- Roles

You must also be familiar with Axon Data Governance, and we recommend that you are familiar with the concepts of IP CIDR and, public and private IP addresses.

Costs and Licenses

You are responsible for the cost of the Google Cloud services while running this deployment. There is no additional cost to use this marketplace deployment.

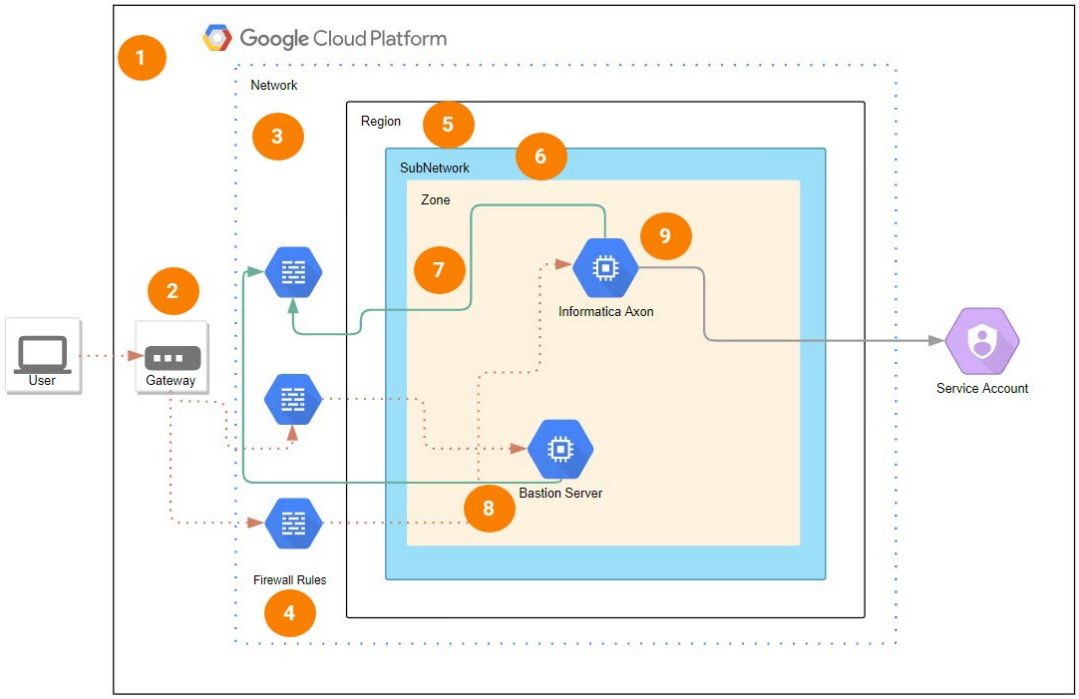
The Google deployment manager template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, affect the cost of deployment. For cost estimates, see the pricing pages for each Google Cloud service that you plan to use. This deployment requires a license for Axon Data Governance. To sign up for a license, contact your organization's Informatica sales contact or Informatica Global Customer Support.

The following table lists the instance types that you can choose based on sizing requirements:

Virtual Machine	Instance Type
Informatica Axon node	n1-standard-8
Bastion server	n1-standard-4

Architecture

The following image shows the architecture of the Axon Data Governance solution on the Google Cloud Marketplace:



1. The Google Cloud Platform (GCP).
2. A gateway to control external access.
3. A network to contain the Axon Data Governance deployment.
4. Firewall rules to control access.
5. Region in which the solution is deployed.
6. Subnets to contain specific elements of the deployment.
7. Zone to contain Axon Data Governance and the bastion server.
8. Bastion server, if you choose to configure one, to serve as an access point for Axon Quick Look.
9. The Informatica Axon Data Governance node.

Google Cloud Platform Resources in the Deployment

When you deploy the Google Cloud Platform (GCP) Marketplace solution on a network, it creates and connects the following resources in the network:

- Informatica Axon Data Governance server on a GCP Compute Engine.
- Informatica Axon Quick Look on a bastion server. (Optional)

Components in a Deployment on an Existing Network

The following components are created when you deploy the software:

Component	Description
Firewall rules	The following firewall rules are created: - One rule for the Informatica Axon Data Governance instance. - One rule to support remote login to the bastion server. - One rule for internal communication.
Virtual Machine	- One GCP Compute Engine running on n1-standard-8 for Informatica Server. - One GCP Compute Engine running on n1-standard-4 for bastion server. (Optional)

Bastion Server

You can optionally deploy a bastion server as an element in the solution. The bastion server acts as a firewall between the internet and the cloud platform network where the solution deploys. It can also act as a remote Windows server that you can log in to run Informatica Axon Quick Look.

Informatica Axon Data Governance

The Axon application consists of the core Axon component, application services and microservices, and databases. These components exchange information through REST APIs.

Application Services

Axon uses application services and microservices to provide specific functionality that enable you to implement data governance. These services are independent, modular services that work together to enhance Axon capabilities.

Informatica® Axon Data Governance includes the following Axon application services:

Authorization

Provides access to Axon objects based on the segments that are assigned to users.

Bulk Upload

Uploads a large number of Axon objects and saves them to the PostgreSQL database.

Change Request

Provides a controlled and audited process to make changes to Axon objects. Axon uses the Camunda service to manage and run workflows within a change request.

Data Marketplace

Provides access to Axon Data Marketplace. In Axon Data Marketplace, you can search and access data collections that Informatica® Axon Data Governance users publish.

Document Upload

Attaches and maintains reference documents to an Axon object.

Notification

Sends notifications about changes to user roles, status of workflows and change requests, and status of bulk upload tasks.

Unison

Performs the Unison search. When you run a Unison search, Axon searches for the objects and the linked objects in all the available facets.

Value List

Creates or updates a large number of values for data set attributes and saves the values to the PostgreSQL database.

Informatica Axon Quick Look

Informatica Axon Quick Look is a client program that you can quickly access from the desktop. Use Axon Quick Look to search and look up glossary terms that are defined in Axon Data Governance. You can also select keywords from applications, such as Microsoft Word, Microsoft Excel, and Google Chrome, and use the hotkeys to perform a lookup in the Axon Quick Look application.

You can optionally deploy a bastion server as an element in the solution. The bastion server is a Windows instance installed with the Axon Quick Look client.

The bastion server has the following functions:

- Acts as a firewall to access the network.
- Acts as a remote Windows server that you can log in to run the Axon Quick Look client.

Note: On the Windows operating system, stop the Quick Look process from the **Task Manager**. After this, change ownership of the `C:\Informatica` directory to the user account that you used to log in to Windows. Now, restart Axon Quick Look.

Prerequisites

Before you launch the automated deployment on the GCP Marketplace, verify the following prerequisites and make the choices described in this section.

License Key Prerequisites

Verify that you have a license to deploy Axon Data Governance.

You supply the license key value in the Informatica Axon License Key parameter when you configure the deployment. Place the Axon Data Governance license key file for the software in the directory that you created.

If you do not have an Axon license, contact Informatica to sign up for a license.

Network Prerequisites

The automated deployment builds a new Google Cloud environment consisting of the firewall rules and other infrastructure components, and then deploys Axon Data Governance into this environment. Before you deploy Axon Data Governance in the Google Cloud, verify that you have completed the network prerequisites for the Informatica application services.

Observe the following prerequisites when you choose to deploy into an existing network:

- If you don't intend to use public IP, verify that Google private access is enabled.
- Verify that the following components are created:
  - Cloud NAT gateway for the deployment zone.
  - If public IP association is not required, select a subnetwork that is in the same zone that includes the Cloud NAT.
- Ensure that the selected network has Cloud NAT on the selected zone while deploying the product without a public IP address.

- The deployment uses the existing network and subnetwork. When you deploy the product without a public IP address, ensure that the selected subnetwork is enabled for Google private access .

To create a new Google Cloud network and subnetwork, perform the following steps:

1. Log in to the Google Cloud console.
2. In the side bar menu, navigate to **VPC network** > **VPC networks** > **CREATE VPC NETWORK**.
3. Enter the details.
4. Click **Create**.

To create a Cloud NAT, perform the following steps:

1. Log in to the Google Cloud console.
2. In the side bar menu, navigate to **Network Services** > **Cloud NAT** > **Get Started**.
3. Enter the details.
4. Click **Create**.

Account and Security Prerequisites

Verify that you have established the required Google Cloud account and have the requisite account private keys.

Google Cloud Account and Region

If you do not have a Google Cloud account, create one.

Use the region selector to choose the GCP Compute Engine region where you want to deploy Axon Data Governance.

If necessary, request a service level increase for the GCP Compute Engine instance type that you plan to use for the Axon Data Governance node. You might need to increase the service level if you already have an existing deployment that uses this instance type, and you might exceed the default limit with this reference deployment.

GCP Compute Engine APIs

Verify that the following APIs are enabled:

- Compute Engine API
- Cloud Deployment Manager V2 API
- Cloud Runtime Configuration API
- Cloud Dataproc API
- Identity and Access Management (IAM) API
- Cloud Resource Manager API
- Google Cloud DNS API
- Service Usage API
- Cloud Storage
- Google Cloud APIs
- Google Cloud Storage JSON API
- IAM Service Account Credentials API

GCP Compute Engine Roles

Verify that the GCP Compute Engine project associated with your deployment has the required roles enabled. You can identify the project with the <project-number>@cloudservices.gserviceaccount.com service account. The <project-number>@cloudservices.gserviceaccount.com service account might be used by the deployment manager for other projects in your account, therefore it requires specific roles.

Ensure that the project that you use has the Owner role, or that the following roles are assigned to the service account :

- Deployment Manager Editor
- Deployment Manager Type Editor
- Project IAM Admin
- Service Account Admin
- Service Account Key Admin
- Compute Admin
- Cloud RuntimeConfig Admin
- Dataproc Administrator
- Dataproc Worker
- Compute Storage Admin
- Storage Admin

Deployment Steps

You use a three-step process to deploy the Axon Data Governance 7.2.3 solution on the Google Cloud Marketplace.

Step 1: Launch the Deployment

Each deployment takes approximately one to two hours to complete.

Perform the following steps to launch the deployment:

1. Use the Google Cloud account to log in to the Google Cloud marketplace.
2. In the search box, type `Informatica Axon Data Governance 7.2.3`, and press **Enter** on your keyboard.

   The **Overview** page appears.
3. Click **Launch**.

   The **Specify Deployment Details** page appears.

Step 2: Specify the Deployment Details

To deploy the Axon Data Governance solution to an existing network, supply the following parameters:

1. On the **Specify Deployment Details** page, enter a name for the deployment. The name can include letters (a-z), numbers (0-9), and dashes (-).
2. Enter values for the following networking parameters:

Parameter	Description
Zone	Required. Zone in which the solution is deployed.
Network	Network that contains the deployment.
Subnetwork	The IP address for the subnetwork. **Note:** This subnetwork must be in the network identified in the **Network** parameter.

Parameter	Description
CIDR IP Range	Required. The CIDR IP range that is permitted to access Axon Data Governance. To reduce the potential of inbound attacks from unknown IP addresses, you can use a constrained CIDR range. For example, to specify the range of 10.20.30.40 to 10.20.30.47, enter 10.20.30.40/29.
Deploy Bastion Server	Select to deploy a bastion server to access other resources in the network. Default is false. **Note:** The Axon Quick Look client is already installed on the bastion server at the time of deployment.
Enable Public IP	Select to deploy compute resources with public IP addresses. Default is true.

3. Enter the following parameters to deploy Informatica Axon Data Governance :

Parameter	Description
Informatica Axon License Key	Required. Axon Data Governance License key value. For more information about how you can secure an Axon license key, see "License Key Prerequisites" on page 6.

4. Confirm that the Google Cloud Platform APIs are enabled.

   Note: To enable the APIs, click the URLs below the **Confirm necessary GCP APIs are enabled** field. After you validate the Cloud APIs, ensure that you select the checkbox next to the **Confirm necessary GCP APIs are enabled** field before you start the deployment.

5. To launch the deployment, click **Deploy**.

Verify the Solution Deployment

After you start the deployment, you can view the current deployment status of each component on the **Output** section of the **Details** page.

Output Properties

On the **Output** tab, you can view the list of resources that the deployment process creates in the network. The following table lists the outputs:

Output	Description
Informatica Axon Administrator URL	The URL with the port for the Axon application. For more information about how you can log in to Axon, see the *Logging In to Axon* topic in the *Axon Data Governance User Guide*.
Informatica Bastion Server IP	IP address of the bastion server. This is available only if you had selected it at the time of deployment

For more information about Axon Data Governance administrator tasks, see the *Axon Data Governance Administrator Guide*.

Log Files

After the completion of the Axon Data Governance deployment, you can view the following logs to see the solution deployment details:

- Command execution log file: `/opt/Infa_OnceClick_Solution.log`
- Axon Configuration log file: `/opt/Informatica/AxonServer/Configure.log`
- Axon Installation log file: `/opt/Informatica/AxonServer/Informatica_Axon_7.2.3.log`

Note: The command execution log records the creation of Informatica Axon Data Governance. If any of the tasks fail to complete, you can look at the detailed section for the task to troubleshoot.

If the above log files are not generated, navigate to the **VM Instances** page. Under **Informatica Axon > Serial port 1 or Serial port 0 (console)**, you can view the VM initialization logs that Google creates for detailed diagnostics.

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